

Name:

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TRANSPORT PHENOMENA II, CHE 312

QUIZ 2

CLOSED BOOK, TIME: 20 MINUTES

A cold liquid film is flowing down a vertical solid wall with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right]$$

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The liquid film, initially at a uniform temperature T_1 comes into contact with a warm gas that maintains the gas-liquid interface temperature at T_0 .

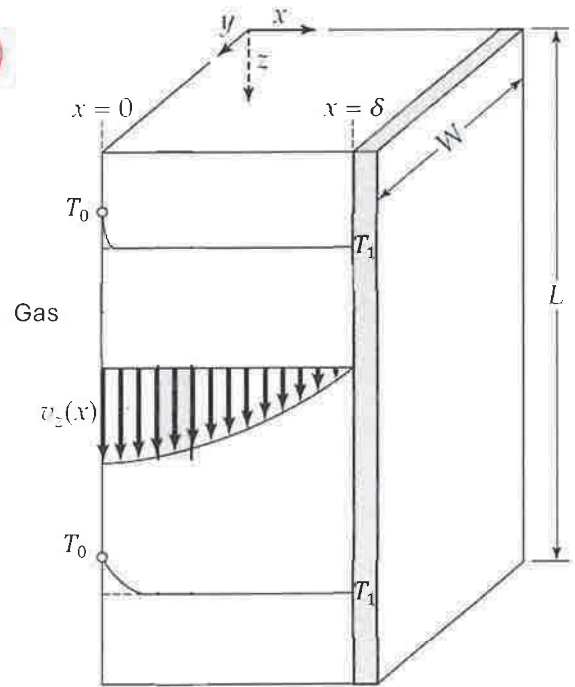
Postulates:

- Liquid has constant density ρ , heat capacity c_p , and thermal conductivity k .
- Short contact times, so the liquid temperature changes appreciably only in the immediate vicinity of the gas interface.
- For $\frac{x}{\delta} \ll 1$ assume $v_z \approx v_{z,max}$

1- Simplify the equation of energy to derive a partial differential equation (PDE) for the steady-state dimensionless temperature $\theta = \frac{T - T_1}{T_0 - T_1}$ as a function of x and z .

2- Convert the above PDE to an ordinary differential equation for θ as a function of $\eta = \frac{x}{\sqrt{4\beta z}}$, where $\beta = \frac{k}{\rho c_p v_{z,max}}$.

3- Write down two boundary conditions for $\theta = \theta(\eta)$.



§B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT^a ρ AND k

$$[\rho \hat{C}_p DT/Dt = k \nabla^2 T + \mu \Phi_v]$$

Cartesian coordinates (x, y, z) :

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-1})^b$$

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$$T_1 \rightarrow T_0$$

assume S.S

assume $v_z = v_{z, \max}$ for $\frac{x}{\delta} \ll 1$

constant ρ, c_p, δ, k

$$v_z = v_{z, \max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right]$$

$$x=0$$

$$1) \theta = \frac{T - T_1}{T_0 - T_1}$$

eq of energy $\left[\rho \hat{c}_p \frac{DT}{Dt} = k \nabla^2 T + \mu \nabla^2 v_z \right]$

constant

$$\theta = \frac{(T_0 - T_1)}{(T_0 - T_1)} \frac{\theta(T_0 - T_1)}{T_0 - T_1} = \frac{T - T_1}{T_0 - T_1} \quad \frac{\rho \hat{c}_p \left(\frac{DT}{Dx} \right)}{\rho \hat{c}_p} = \frac{k \nabla^2 T}{\rho \hat{c}_p}$$

$$\frac{DT}{Dx} = \frac{k \nabla^2 T}{\rho \hat{c}_p} \Rightarrow \frac{DT}{Dx} = \frac{k \nabla^2 (T_1 + \theta(T_0 - T_1))}{\rho \hat{c}_p}$$

$$X = \frac{\partial T}{\partial z} \left(\frac{k u}{\rho^2 g \delta c_p} \right) \frac{\partial^2 T}{\partial y^2} =$$

$$X = \frac{\partial T}{\partial z} \left(\frac{u}{\rho^2 g \delta} \right) \left(\frac{\partial^2 (T_1 + \theta(T_0 - T_1))}{\partial y^2} \right)$$

$$2) X = \frac{\partial T}{\partial z} \left(\frac{u}{\rho^2 g \delta} \right) \left(\frac{\partial^2 (T_1 + \theta(T_0 - T_1))}{\partial y^2} \right) = \frac{\partial T}{\partial z} \left(\frac{u}{\rho^2 g \delta} \right) \left(\frac{\partial^2 T}{\partial y^2} \right) = X$$

$$3) v_x(T_1) = T_0$$

$$v_z(\theta) = T_1$$

1) } incorrect
2) }
3) }